Universidade Leiria • Oeste
Escola de Tecnologia e Gestão

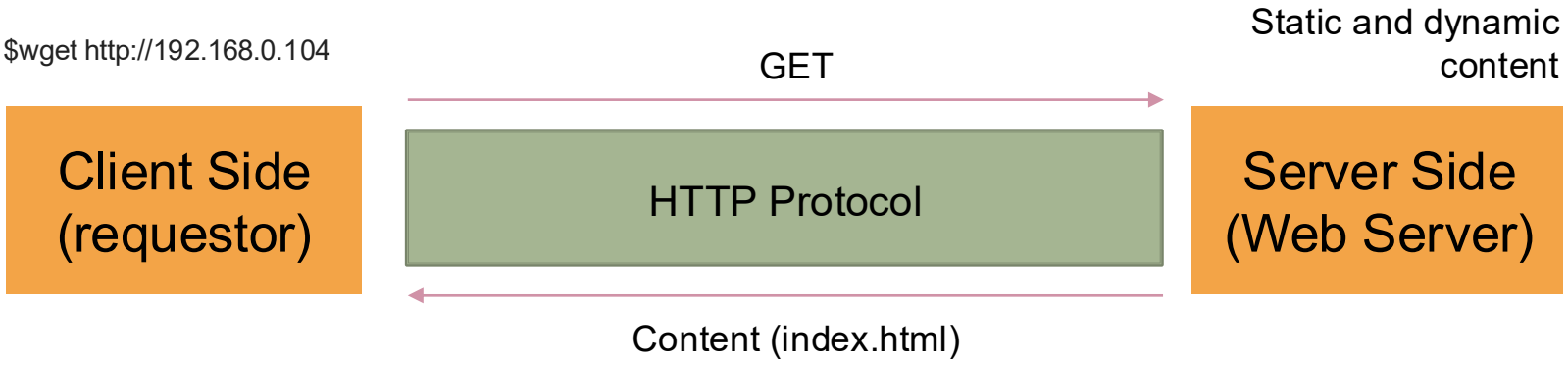
Information Systems Integration

Part 5 – Web Services:
SOAP, WSDL and UDDI

Background

After Web Services and SOA introduction…

The Web Big Picture



Background

- After Web Services and SOA introduction…

Web Services= XML + Web Standards



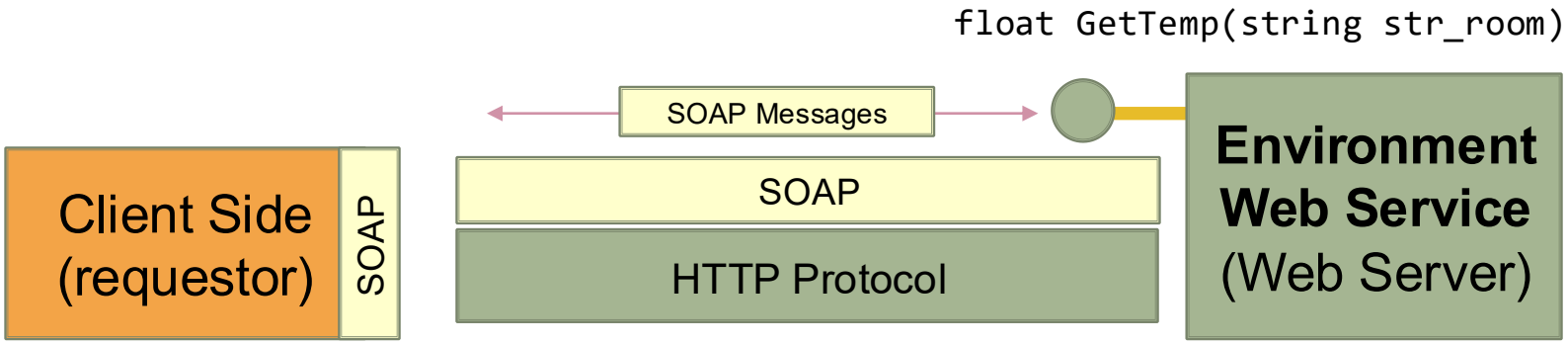
1. How to call the method?
(It is not a web page like in last slide scenario)
(wget can not do that)

?

Background

- After Web Services and SOA introduction…

Web Services= XML + Web Standards



Background

- After Web Services and SOA introduction…

Web Services= XML + Web Standards

How do we know about the GetTemp() method's existence?



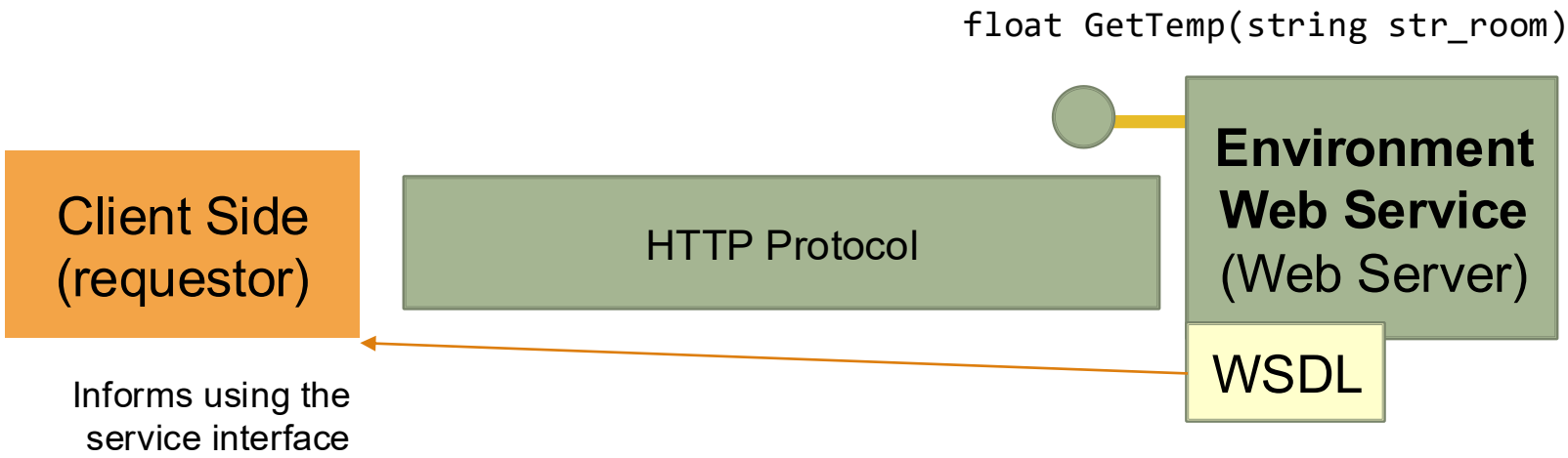
```
float GetTemp(string str_room)
```



Background

- After Web Services and SOA introduction…

Web Services= XML + Web Standards



Simple Object Access » Protocol (SOAP)

Web Services and SOAP

- SOAP - Simple Object Access Protocol
  - It's an **RPC** mechanism
  - It's an interoperability standard
  - It's XML based

SOAP = RPC over HTTP[1]

[1] Not only HTTP

Web Services and SOAP

- SOAP - Simple Object Access Protocol
  - SOAP XML elements
    - **Envelope – defines the message (mandatory)**
      - Header – optional attributes (optional)
      - **Body – method and arguments (one, mandatory)**
        - Fault – info about errors while processing message (opt)

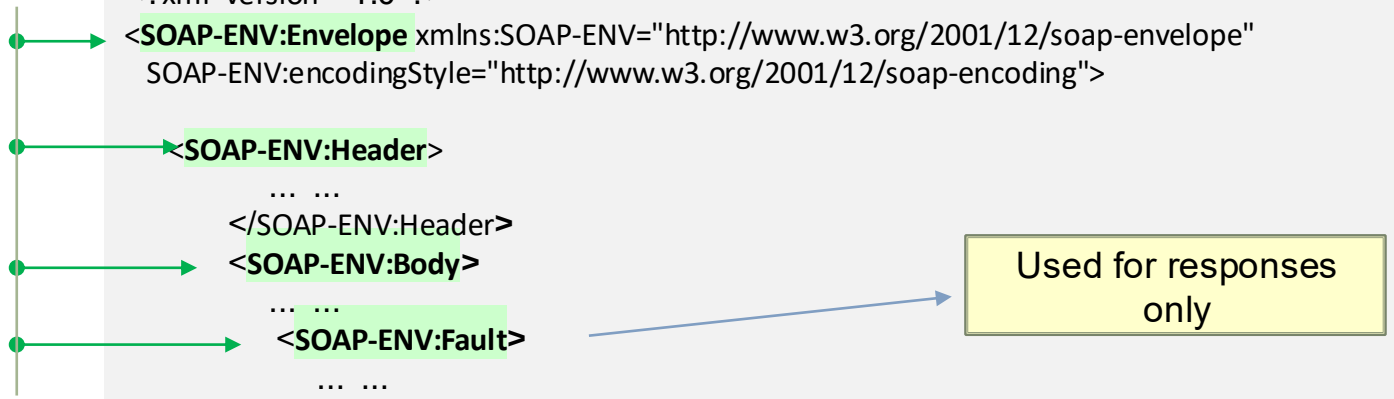
Important namespaces:

http://www.w3.org/2001/12/soap-envelope - structure
http://www.w3.org/2001/12/soap-encoding - datatypes, encoding

Web Services and SOAP

SOAP - Simple Object Access Protocol

SOAP message example



Web Services and SOAP

SOAP - Simple Object Access Protocol

SOAP message example in an HTTP Post

```
POST /OrderEntry HTTP/1.1
Host: www.abc.com
Content-Type: application/soap; charset="utf-8"
Content-Length: nnnn

<?xml version="1.0"?>

<SOAP-ENV:Envelope xmlns:SOAP-ENV="http://www.w3.org/2001/12/soap-envelope"
  SOAP-ENV:encodingStyle=" http://www.w3.org/2001/12/soap-encoding">
   ...
   Message information goes here
   ...
</SOAP-ENV:Envelope>
```

```
GET /docs/index.html HTTP/1.1
Host: www.test.com
Accept: image/gif, image/jpeg, */*
Accept-Language: en-us
Accept-Encoding: gzip, deflate User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)
(blank line)
```

Web Services and SOAP

- SOAP - Simple Object Access Protocol
  - **SOAP Header (optional)**
    - Application-level details (previously agreed)
    - Must understand attribute
      - If present, the other peer must interpret element or return a fault

```
<SOAP-ENV:Header>
      <t:SecretKey xmlns:t="http://www.mydomain/sec/" SOAP_ENV:mustUnderstand="true“ >#”$#$%%&$GRD#$%</t:SecretKey>
</SOAP-ENV:Header>
```

Web Services and SOAP

- SOAP - Simple Object Access Protocol
  - **SOAP Body (mandatory)**
    - Mandatory information needed by the recipient

```
<?xml version="1.0"?>
<SOAP-ENV:Envelope ...>
  <SOAP-ENV:Body>

    <m:GetPrice xmlns:m="http://www.mydomain.com/Price">
          <m:Item>Laptop</m:Item>
      </m:GetPrice>

  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

Application specific

A request…

Web Services and SOAP

- SOAP - Simple Object Access Protocol
  - SOAP Body (mandatory)
    - Mandatory information needed by the recipient

A response…

```
<?xml version="1.0"?>
<SOAP-ENV:Envelope ...>
  <SOAP-ENV:Body>

    <m:GetPriceResponse xmlns:m="http://www.xyz.com/Price">
          <m:Price>600.00</m:Price>
    </m:GetPriceResponse>

  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

Application specific

SOAP processor is responsible for mapping elements into app methods

Web Services and SOAP

SOAP - Simple Object Access Protocol

SOAP Fault (if some error takes place during XML processing)

```
<?xml version='1.0' encoding='UTF-8'?>
<SOAP-ENV:Envelope
    xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/"
    xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/1999/XMLSchema">

  <SOAP-ENV:Body>

    <SOAP-ENV:Fault>
      <faultcode xsi:type="xsd:string">SOAP-ENV:Client</faultcode>
        <faultstring xsi:type="xsd:string">
          Failed to locate method (GetPrice) in class (AllPrices)
            </faultstring>
        </SOAP-ENV:Fault>
  </SOAP-ENV:Body>

</SOAP-ENV:Envelope>
```

Ok, all this is XML…
What is missing in this request/response process?



Web Services and SOAP

SOAP - Simple Object Access Protocol

SOAP Fault (if some error occurs during XML processing)

```
<?xml version='1.0' encoding='UTF-8'?>
<SOAP-ENV:Envelope
    xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/"
    xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/1999/XMLSchema">

  <SOAP-ENV:Body>

    <SOAP-ENV:Fault>
      <faultcode xsi:type="xsd:string">SOAP-ENV:Client</faultcode>
        <faultstring xsi:type="xsd:string">
          Failed to locate method (GetPrice) in class (AllPrices)
            </faultstring>
        </SOAP-ENV:Fault>
  </SOAP-ENV:Body>

</SOAP-ENV:Envelope>
```

Ok, all this is XML…
What is missing in this request/response process?
The interface!

Web Services Description Language (WSDL)



Web Services and WSDL

WSDL - Web Services Description Language

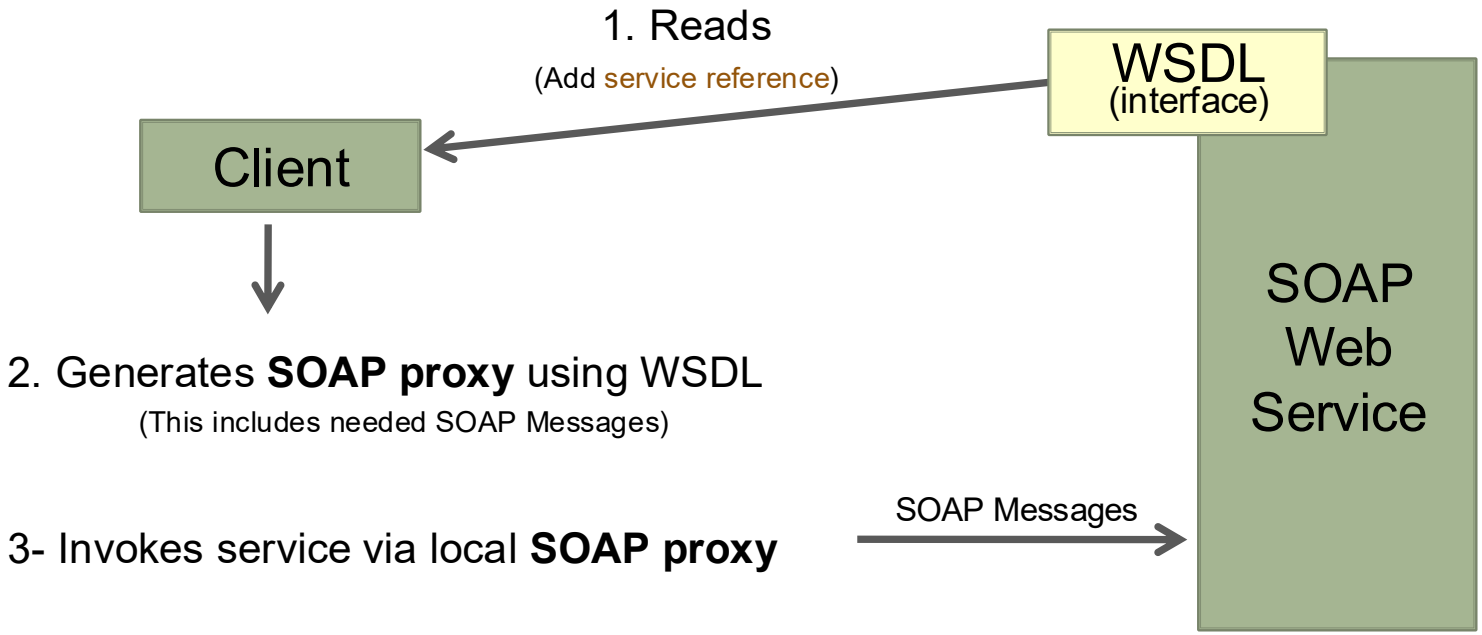
- Describes a web service (XMLschema-based)
- A client reads WSDL…
- Any special datatypes used are embedded…
- The client uses **SOAP** to actually call functions

Tools can create WSDL from implementation or implementation from WSDL

Web Services and WSDL

WSDL - Web Services Description Language

Hence:



Web Services and WSDL

WSDL - Web Services Description Language

WSDL Elements (just for curiosity)

Definition	– root element
Data types	– user defined data types
Message	– arguments
Operation	– methods
Port type	– endpoint and binding
Binding	- protocol
Port	– address
Service	– map binding with the port

Web Services and WSDL

WSDL - Web Services Description Language

WSDL example (just for curiosity)

```
<definitions name="HelloService"
  targetNamespace="http://www.examples.com/wsdl/HelloService.wsdl"
  xmlns="http://schemas.xmlsoap.org/wsdl/"
  xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/"
  xmlns:tns="http://www.examples.com/wsdl/HelloService.wsdl"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">

  <message name="SayHelloRequest">
    <part name="firstName" type="xsd:string"/>
   </message>

  <message name="SayHelloResponse">
    <part name="greeting" type="xsd:string"/>
   </message>
```

Web Services and WSDL

WSDL - Web Services Description Language

WSDL example (just for curiosity)

```
<portType name="Hello_PortType">
      <operation name="sayHello">
        <input message="tns:SayHelloRequest"/>
        <output message="tns:SayHelloResponse"/>
        </operation>
</portType>
```

Web Services and WSDL

WSDL - Web Services Description Language

WSDL example (just for curiosity)

```
<binding name="Hello_Binding" type="tns:Hello_PortType">
   <soap:binding style="rpc"
     transport="http://schemas.xmlsoap.org/soap/http"/>
     <operation name="sayHello">
      <soap:operation soapAction="sayHello"/>
            <input>
                <soap:body
          encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
          namespace="urn:examples:helloservice"
                    use="encoded"/>
            </input>

            <output>
              <soap:body
          encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
          namespace="urn:examples:helloservice"
                    use="encoded"/>
            </output>
        </operation>
   </binding>
```

Web Services and WSDL

WSDL - Web Services Description Language

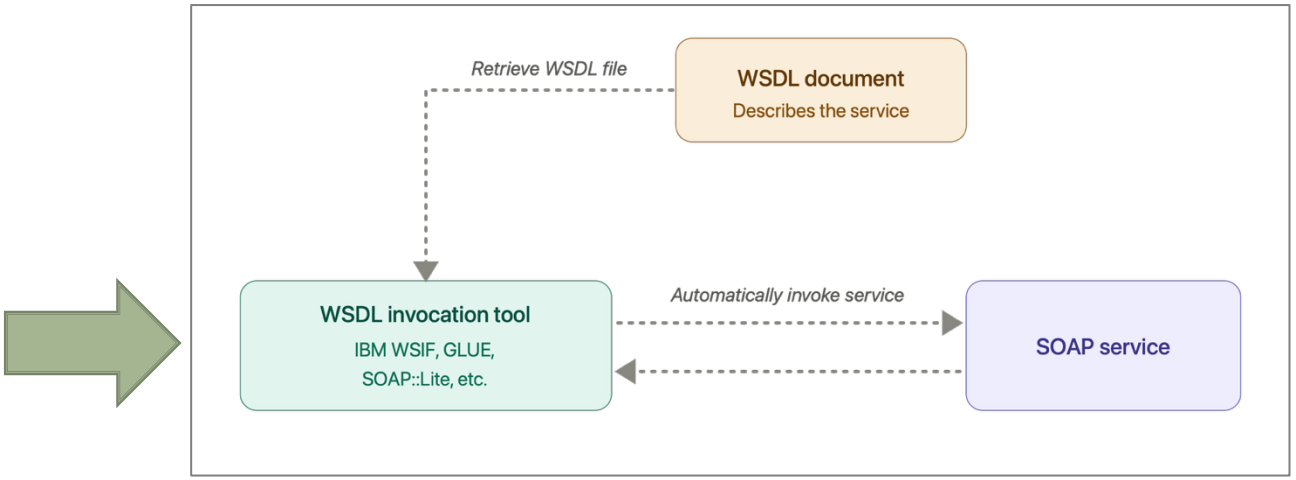
WSDL example (just for curiosity)

```
...
<service name="Hello_Service">
   <documentation>WSDL File for HelloService</documentation>
   <port binding="tns:Hello_Binding" name="Hello_Port">
      <soap:address
        location="http://www.examples.com/SayHello/" />
   </port>
</service>
</definitions>
```

Web Services and WSDL

WSDL - Web Services Description Language

HelloService.wsdl – (Automatic) Invocation tools



WCF Test Client → is a Microsoft tool that uses the WSDL to automatically generate a SOAP client, allowing developers to test Web Service operations without writing client-side code

Finding Web Services: UDDI

- Universal Description, Discovery, and Integration (Registry)

Background

UDDI - Universal Description, Discovery, and Integration Registry

- XML-based standard for advertising and finding web services
  - Includes business and service information
- It is a **directory** of Web Services (Spec)
- It is also a **Web Service** by itself ☺

UDDI supports replication (works similarly to DNS)

? Alternative method?

Background

UDDI - Universal Description, Discovery, and Integration Registry

UDDI is a directory service that allows organizations to **publish**, **discover**, and **locate** SOAP Web Services

Think of UDDI as a **"phone book for web services"**

- **Service Providers** publish information about their services
- **Service Consumers** search for available services
- The UDDI registry returns service details, including the location of the **WSDL** document
- Consumers use the WSDL to understand how to communicate with the service

Background

UDDI - Universal Description, Discovery, and Integration Registry

How it Works

1. **Publish** → A provider registers a SOAP service in the UDDI registry
2. **Discover** → A client searches for a service by name, category, or capability
3. **Retrieve** → The client obtains the service's WSDL and endpoint information
4. **Invoke** → The client sends SOAP requests to the service

Background

UDDI - Universal Description, Discovery, and Integration Registry

Two types of APIs

UDDI Client for service registering or SOAP

1. Get Access Token…

Publishing API

save_business
save_service
save_binding
save_tModel
delete_business
…

UDDI Client for service searching or SOAP

No access token needed

Inquiry API

find_business
find_service
find_binding
find_tModel
find_businessDetail
…

UDDI Registry provided by a UDDI operator

Background

UDDI - Universal Description, Discovery, and Integration Registry

Example of UDDI registries:

Apache jUDDI: Operator node and Test Bed UDDI

http://uddi-jbossoverlord.rhcloud.com/

http://uddiconsole-jbossoverlord.rhcloud.com (testing version)

(You can just download jUDDI and run it locally as well)

By the way…

Microsoft, IBM and SAP shut down their public UDDI back in 2007…

References

- [1] Ethan Cerami, “Web Services Essentials: Distributed Applications with XML-RPC, SOAP, UDDI & WSDL”, O’REILLY, 2002 [Chapter 6]

- [2] Eric Newcomer, “Understanding Web Services: XML, WSDL, SOAP and UDDI”, Addison-Wesley, 2002 [Chapter 5]